Unit 2 – Selection and Iteration Name: **KEY**
HW 4

1) For the code segments below, trace through the loops and determine what the value of `sum` will be after the code has executed. SHOW YOUR WORK!

```
int value = 3;
int sum = 0;
while(value > 0)
{
      sum++;
      value--;
}
```

value	sum
~~3~~	~~0~~
~~2~~	~~1~~
~~1~~	~~2~~
0	3

```
int value = 2;
int sum = 0;
while(value % 10 != 0)
{
      sum++;
      value += 2;
}
```

value	sum
~~2~~	~~0~~
~~4~~	~~1~~
~~6~~	~~2~~
~~8~~	~~3~~
10	4

```
int value = -10;
int sum = 0;
while(value + 10 < 5)
{
      sum++;
      value += sum;
}
```

value	sum
~~-10~~	~~0~~
~~-9~~	~~1~~
~~-7~~	~~2~~
-4	3

2) How many times will each loop below execute?

```
int i = 1;
while(i < 100)
{
      i++;
}
```

99 times

```
int i = 5;
while(i <= 100)
{
      i++;
}
```

96 times

```
int i = 5;
while(i < 125)
{
      i++;
}
```

120 times

3) Write a code segment that does the following with the integer `value`.

```
int value = /* not shown */;
```

a) Sums every integer from 0 up to and including `value` and store it in a variable called `sum`.

```
int sum = 0;
while(value > 0)
{
     sum += value;
     value--;
}
```

b) Find the factorial of the value and store it in a variable called `fact`.

```
int fact = 1;
while(value > 0)
{
     fact *= value;
     value--;
}
```

A 3) Consider the following code:

```
int count = 4;
while(count <= 7)
{
  count++;
  System.out.print(count + " ");
}
```

What are the first and last numbers that are printed?

(A) 5 8
(B) 4 7
(C) 5 7
(D) 4 8

D 4) How many times will the following loop repeat?

```
int num = 49;
while(num > 0)
{
  if(num % 2 == 0)
  {
    num++;
  }
  else
  {
    num--;
  }
}
```

(A) 21 (B) 22 (C) 23 (D) This is an infinite loop

B 5) Consider the following code:

```
int n = 14;
while(!(n % 3 == 0 && n % 5 == 0))
{
  System.out.println(n);
  n += 2;
}
```

What is the last number printed by this code?

(A) 16 (B) 28 (C) 18 (D) 30

While n is not evenly divisible by 3 and 5, keep adding 2 to it

n = 14, 16, 18, 20, 22, 24, 26, 28, 30

Since 30 is the first number divisible by 3 and 5, the loop ends after 28 is printed.

D 6) Consider the following code:

```
int a = 0;
int b = 0;
while(a < 5 && b < 3)
{
  System.out.println(a + " " + b);
  a++;
  b++;
}
```

What is the output?

(A)	(B)	(C)	(D)
0 0	0 0	0 0	0 0
1 1	1 1	1 1	1 1
2 2	2 2	2 2	2 2
3 3	3 3	3	
4 4		4	
5 5			

A 7) The following piece of code intends to count how many integers a user enters until they enter a "–1". The program will then print off how many integers they entered.

```
System.out.print("Enter integers. Enter -1 to exit.");
int numEntered = 0;
int count = 0;
while(numEntered != -1)
{
  numEntered = scan.nextInt();
  count++;
}
System.out.println(count);
```

What, if anything, causes the code to not work as intended?

(A) The loop counts the value –1 when it is entered, so the count will be one too many.
(B) The code will not compile.
(C) The condition for ending the loop should use `count` not `numEntered`.
(D) The loop counts the value –1 when it is entered, so the count will be one too few.